

Project Abstract

Project Title:

Supermarket Billing System

Abstract:

The project Supermarket Billing System is developed to make the billing process in a supermarket easier and faster. The main objective of this project is to calculate the bill accurately and reduce the time taken for billing. The proposed system provides features such as adding products, entering quantities, calculating the total amount, and generating the bill. It is developed using programming and database tools. The system is designed to make billing simple, reduce calculation mistakes, and keep product details organized. This project aims to provide an easy, fast, and reliable solution for supermarket billing.

In addition to the core billing functions, the system also includes a Smart Expiry Management feature that helps reduce expired-product waste, and an Automatic Discount System that applies discounts based on expiry, quantity, or ongoing offers. A Low-Stock Alert notifies staff when products need restocking, while a Digital Receipt feature sends or creates a paperless receipt for customers. The system also maintains a Customer Purchase History, keeping previous purchases on record for easier future billing, and supports Barcode/QR Scanning to automatically identify products at checkout. A built-in Sales Analysis feature shows best-selling and slow-moving products to support better inventory decisions, and Voice-Assisted Billing allows product and quantity input through voice commands, making the billing process even more accessible and efficient.

Keywords:

Supermarket, Billing, Products, Bill Generation, Database, Expiry Management, Discounts, Stock Alerts, Digital Receipt, Barcode Scanning, Sales Analysis, Voice Billing

Supermarket Billing System

Feature	SDG No.	SDG Name	Detailed Explanation
 Digital Billing & Bill Generation	9	Industry, Innovation and Infrastructure	The system replaces manual billing with a digital billing process. It makes bill preparation faster, more accurate, and reduces calculation errors.
 Paperless Billing	12	Responsible Consumption and Production	Digital bills reduce the use of paper and help avoid unnecessary printing. This supports responsible use of resources.
 Faster & Efficient Billing	8	Decent Work and Economic Growth	The system reduces the time and effort required for billing. It helps supermarket staff handle customers more efficiently and improves overall productivity.
 Easy & Affordable Shopping	10	Reduced Inequalities	The system provides a simple billing process for all customers and clearly shows product prices and the total amount, making the payment process more transparent.